DAT301m Final Report - Floating Human Detection Using Drone

Group 7 - AI1803

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Abstract

This study proposes an optimized framework for detecting floating humans at sea (serving Search and Rescue - SAR purposes) using Unmanned Aerial Vehicle (UAV) imagery. Based on the SeaDronesSee dataset [1], we observed that traditional object detection methods often struggle with extremely small objects and visually noisy environments. Therefore, this research utilizes the YOLOv8-p2 [2] architecture [3] specifically tailored for high-resolution feature extraction. To address the severe scale variations and prevent the loss of fine-grained details inherent in drone-viewpoint imagery, we integrate Slicing Aided Hyper Inference (SAHI) [4].

1 Introduction

In recent years, the deployment of Unmanned Aerial Vehicles (UAVs or drones) in maritime Search and Rescue (SAR) missions has become increasingly vital due to their flexibility, broad field of view, and ability to rapidly access hazardous areas. However, automatically detecting humans or floating objects from a high-altitude UAV perspective presents substantial challenges. Targets are typically extremely small relative to the entire image frame and are easily obscured or confused by ocean waves and dynamically changing illumination.

Although modern object detection models like the YOLO series have achieved remarkable success, standard configurations often discard or degrade high-resolution spatial features during downsampling, leading to difficulties in recognizing tiny objects. To address this issue, our report focuses on applying the YOLOv8-p2 architecture. The primary objective is to optimize inference speed to ensure the feasibility of deploying the model directly on hardware-constrained edge devices.

2 Related Works

Object Detection via UAV Imagery: Numerous studies have applied Convolutional Neural Networks (CNNs) and YOLO-based models to analyze UAV imagery. Due to the top-down perspective and variations in flight altitude, these algorithms require robust feature fusion techniques to handle the significant scale variations of objects in real-world contexts. To facilitate this, Varga et al. introduced the SeaDronesSee benchmark to specifically address the gap in large-scale, high-altitude datasets for maritime SAR applications, providing a standardized foundation for evaluating small object detectors over water.

Small Target Detection Enhancements: To improve detection performance for tiny objects, research has shifted toward preserving spatial resolution through architectural and inferencing optimizations. Standard object detection models often begin feature extraction at higher strides (e.g., stride 8 or 16), which effectively erases small targets that occupy only a few pixels. The usage of the P2 detection head, which operates at a stride of 4, allows the network to maintain high-resolution feature maps essential for identifying humans in vast maritime environments.

3 Methodology

This section details the proposed framework for maritime human detection using UAV-perspective imagery. The approach focuses on optimizing small object detection through architectural enhancements and comparing various inference engines for real-time viability.

3.1 Dataset and Preprocessing

The study utilizes the *SeaDronesSee* dataset, which contains high-altitude maritime images focusing on Search and Rescue (SAR) scenarios. We chose the Multi-Object Tracking dataset, which contains 13 videos split into 60% for training, 20% for validation, and for testing. The input consists of video frames, each cut into 4 640×640 pixels using Slicing Aided Hyper Inference (SAHI) matching with YOLOv8 input size. Standard normalization and augmentation techniques were applied during the training phase to improve model generalization across varying sea states.

3.2 Model Architectures

We utilize the YOLOv8-p2 architecture (Fig. 1). Unlike the standard YOLOv8 which begins feature extraction at a stride of 8, the P2 variant introduces an additional detection head at a stride of 4. This preserves high-resolution spatial information, which is critical for objects occupying fewer than 16×16 pixels in the maritime expanse.

3.3 Slicing Aided Hyper Inference (SAHI)

To address the challenge of detecting tiny objects in high-resolution maritime imagery, we integrate Slicing Aided Hyper Inference (SAHI). In typical Search and Rescue (SAR) scenarios, drone-captured images possess high pixel densities, but targets often occupy a negligible fraction of the total area. Standard preprocessing that resizes these large frames to 640×640 pixels results in a significant loss of spatial information, making small targets difficult to recognize.

The SAHI framework overcomes this by cutting the high-resolution input into multiple overlapping patches or slices (e.g., 640×640) during inference. Each slice is fed independently into the YOLOv8-p2 detector. By performing detection on these smaller images, the model preserves fine-grained

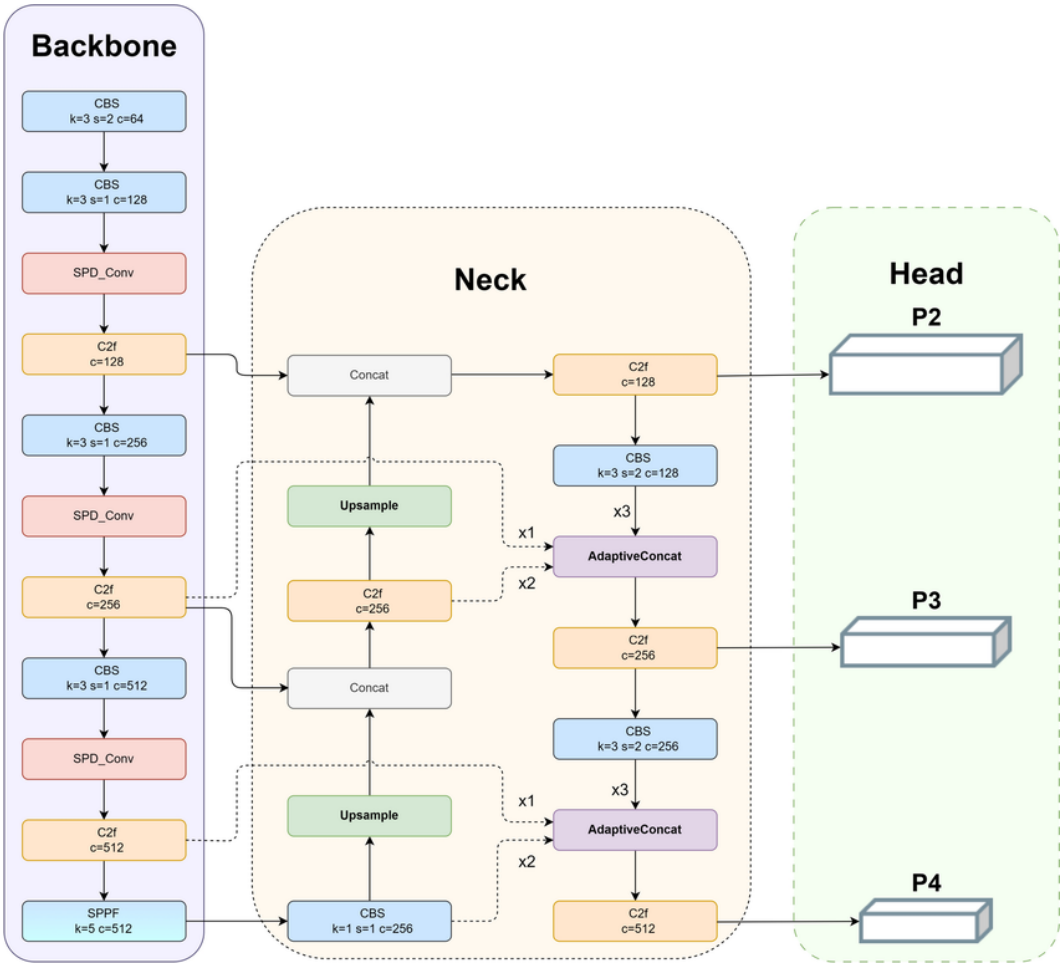


Figure 1: YOLOv8-p2 architecture

features that would be lost during global downsampling. Finally, the individual detection results from each slice are mapped back to the original image coordinates to produce the complete detection output.

3.4 Object Tracking with *ByteTrack*

To ensure temporal consistency in the live video stream, the detection output is coupled with the *ByteTrack* [5] algorithm. Unlike traditional trackers that discard low-score detection boxes, *ByteTrack* leverages the similarity between low-score boxes and existing tracklets to recover objects partially occluded by waves or sea spray. This reduces the False Negative rate in "man overboard" scenarios.

3.5 Model Optimization and Export

We compare (`.pt`) file against (`.tflite`) file by the average inference time by taking the average time for each to process one frame.

4 Experiments and Results

4.1 Experimental Setup

The models were trained using the PyTorch [6] framework on an NVIDIA RTX 4050 GPU. The training process utilized the SGD optimizer with an initial learning rate of 0.01 matching YOLOv8 training, a batch size of 16, and was conducted over 100 epochs. For inference benchmarking, the CPU evaluations for the (`.tflite`) models were performed on an Intel I7 13620H.

4.2 Evaluation Metrics

To quantitatively evaluate detection performance, we employ standard object detection metrics: Precision (P), Recall (R), F1-Score ($F1$), and mean Average Precision (mAP) [7]. Specifically, we report mAP_{50} (calculated at an IoU threshold of 0.5) and mAP_{50-95} (the average mAP calculated across IoU thresholds from 0.5 to 0.95).

Precision and Recall:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\mathrm{P} \cdot \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{1}$$

Average Precision (AP):

$$AP = \int_0^1 p(r)dr \tag{2}$$

Mean Average Precision (mAP):

$$mAP_{50} = \frac{1}{n} \sum_{i=1}^{n} AP_{i,@0.5}, \quad mAP_{50-95} = \frac{1}{10} \sum_{k \in \{0.5, 0.55, \ldots, 0.95\}} mAP_k \tag{3}$$

4.3 Experimental Results

The training process was monitored over 60 epochs to ensure model stability. As shown in (Fig. 2), both the box and classification loss for training and validation sets decreased consistently, reaching a plateau around epoch 40. This convergence indicates that the YOLOv8-p2 model successfully learned the features of the SeaDronesSee dataset without significant overfitting.

The quantitative performance is further detailed by the Precision-Recall (PR) and F1-Confidence curves in (Fig. 3). The model achieved an mAP_{50} of 0.874 for the swimmer class. The F1-curve reaches its peak at a confidence threshold of 0.334, suggesting an optimal balance between precision and recall for maritime search and rescue missions where missing a target (False Negative) is more critical than a False Positive.

Qualitative results are presented in (Fig. 4), comparing the ground truth annotations with the model's predictions. Even in frames with high target density and varying light conditions, the integration of the P2 head and SAHI slicing allowed the model to detect swimmers that occupy a very small pixel area, matching the ground truth labels with high confidence scores.

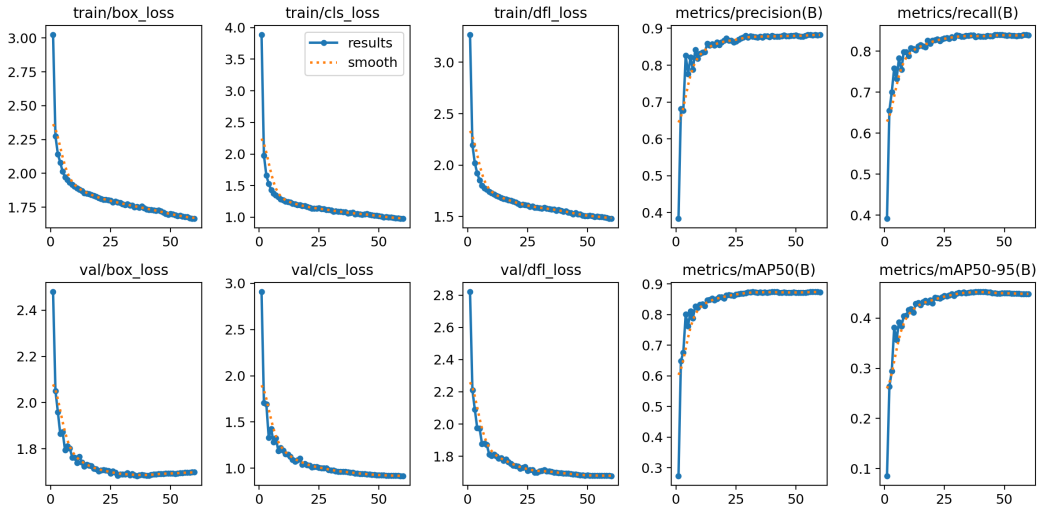
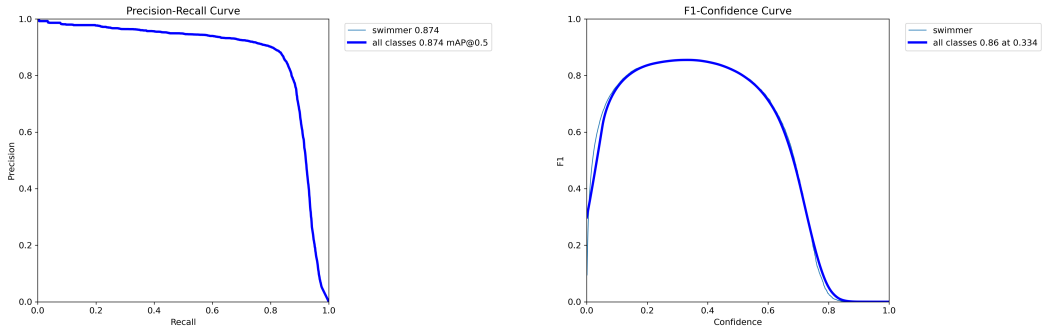


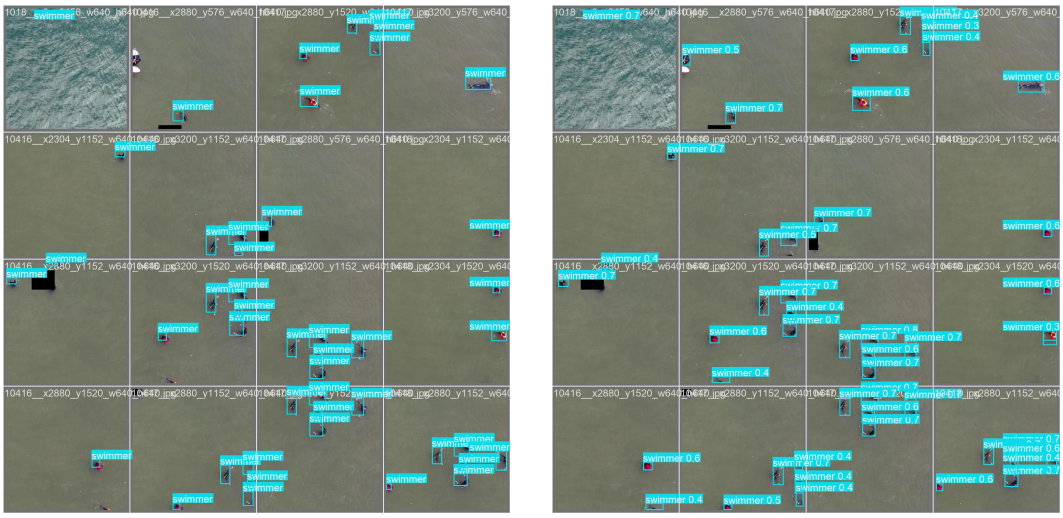
Figure 2: Training and validation loss curves alongside accuracy metrics over 60 epochs.



(a) Precision-Recall Curve

(b) F1-Confidence Curve

Figure 3: Detailed performance metrics for the swimmer class, showing an mAP_{50}.



(a) Ground Truth Labels

(b) Model Predictions

Figure 4: Visual comparison between ground truth and model inference on validation batches.

The model's performance on the maritime SAR dataset is summarized in Table 1.

Table 1: Model Performance Summary

Metric	**Precision**	**Recall**	**F1-score**	mAP_{50}	mAP_{50-95}
Value	0.8790	0.8363	0.8571	0.8745	0.4531

4.4 Inference Speed and Edge Optimization

To address the hardware constraints of UAV deployments, we compared the CPU inference speeds of the standard PyTorch weights against the optimized TensorFlow Lite format. Table 2 details the latency and FPS tradeoffs, proving the model's viability for real-time edge processing.

Table 2: Performance comparison

File	**Average Inference Time**	**Estimated FPS**
`(.pt)`	0.1382	7.24
`(.tflite)`	**0.0422**	**23.69**

Metric Definitions:

- Average Inference Latency: The time, in seconds (s), required for the model to process a single image frame.
- Estimated FPS: The number of frames the system can process in one second.

5 Conclusion

This study developed and evaluated a maritime floating object detection framework tailored for UAV-assisted search and rescue operations. By employing the YOLOv8-p2 configuration and integrating Slicing Aided Hyper Inference (SAHI), the system addressed the limitations of traditional detection methods when dealing with extremely small targets in high-resolution images from the SeaDronesSee dataset. The use of an additional high-resolution detection head, combined with full-resolution image tiling, ensures that critical spatial details of humans at sea are preserved rather than lost during standard image resizing.

Experimental results demonstrate that this approach delivers reliable detection performance in complex maritime environments. Notably, the conversion of model weights to the optimized TensorFlow Lite format proved the system's viability for practical deployment on hardware-constrained edge processors. Consequently, the system achieves an effective balance between Frames Per Second (FPS) and inference latency, satisfying the real-time requirements necessary for life-saving search and rescue applications.

References

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